

Bias Detection Analysis

Upload a financial LLM research paper to automatically evaluate it against the **Structural Validity Framework** — 5 sections and 12 checklist items covering look-ahead bias, survivorship bias, narrative bias, objective bias, and cost bias, with evidence-backed verdicts and actionable recommendations.

MODEL

Provider

Model

API Key

Anthropic

claude-sonnet-4-6

.....

Find model IDs: Anthropic · OpenAI · Google · Ollama

PAPER

Choose a file or drag it here

PDF format

GitHub Repository URL (optional, for code analysis)

https://github.com/user/repo

If provided, the system will also analyze the source code for logging, data loading, cost calculations, etc.

Analyze Paper

THE FIVE SINS CHECKED

Look-Ahead

Temporal Sanitation

Survivorship

Dynamic Universe

Narrative

Rationality Robustness

Objective

Epistemic Calibration

Cost

Realistic Constraints

Evaluating all 12 checklist items... 2/7

Extract

Classify

Look-Ahead
Temporal Sanitation

Survivorship
Dynamic Universe

Narrative
Rationality Robustness

Objective
Epistemic Calibration

Cost
Realistic Implementation

Case 1: Trading Agent

Can LLM-based Financial Investing Strategies Outperform the Market in Long Run?

benchmark · anthropic/claude-opus-4-6 · 30/03/2026, 19:42:58

LLMExternal DataBacktest2004-2024GPT-4o-miniGPT-4oFinMemFinAgent

FinGPTFinConARIMAXGBoostA2CPPOTD3SAC

Daily stock prices for 7000+ US equities (2000-2024)Financial news dataset (15.7 million records, 1999-2023)

10-K and 10-Q filings from SEC EDGARS&P 500 historical constituent lists including delisted symbols

> Paper Classification Details

1. Temporal Sanitation Non-Anticipativity

Parametric Knowledge Cutoff Disclosure

Are the latest dates of pre-training and fine-tuning disclosed? Does the evaluation window strictly follow these dates? X FAIL

The paper uses GPT-4o-mini and GPT-4o (as seen in Table 8: 'FinMem model=gpt-4o-mini' and 'FinAgent model=gpt-4o-mini', plus 'FinMem (GPT-4o)' results in Table 2) for LLM-based strategies evaluated over 2004-2024. However, the paper does not disclose the pre-training or fine-tuning knowledge cutoff dates for these models. The paper actually acknowledges this issue in passing: 'our reported LLM performances do not adjust for potential data leakage: given the use of pretrained models like GPT-4o, the LLMs may have seen parts of the data during training' (Section 6.2). This is a critical admission — the evaluation window (2004-2024) almost certainly overlaps with the training data of GPT-4o/GPT-4o-mini, yet no formal cutoff dates are disclosed and no temporal separation is enforced. The code (e.g., strategy config files and run_llm_traders_exp.py) confirms the use of these models but contains no model version pinning or cutoff date specification.

EVIDENCE

- "Table 8: 'FinMem model=gpt-4o-mini, top_k=3, embedding_model=text-embedding-ada-002'"
- "Table 8: 'FinAgent model=gpt-4o-mini'"
- "Section 6.2: 'our reported LLM performances do not adjust for potential data leakage: given the use of pretrained models like GPT-4o, the LLMs may have seen parts of the data during training, but they still fail to outperform traditional strategies under fair evaluation'"
- "Code: run_llm_traders_exp.py uses FinMemStrategy and FinAgentStrategy with config paths but no model version pinning"

RECOMMENDATIONS

Explicitly disclose the pre-training knowledge cutoff dates for GPT-4o-mini and GPT-4o.

Discuss the temporal overlap between the evaluation window (2004-2024) and the models' training data in a dedicated section.

Consider restricting the evaluation window to periods after the models' knowledge cutoff, or at minimum, clearly flag results from overlapping periods as potentially contaminated.

Pin specific model versions (with snapshot dates) in both the paper and code configuration.

Point-in-Time (PIT) Retrieval

Are retrieval sources constructed using ONLY documents available at the specific decision time? (No live web search or mutable wikis).

✓ PASS

The paper uses external time-varying data including financial news and 10-K/10-Q filings. The paper explicitly states: 'To prevent look-ahead bias, all data inputs are aligned with each backtest window using only information available prior to the start date' (Section 5). The news dataset is a pre-compiled historical dataset from Dong et al. (15.7 million records, 1999-2023), organized by company and indexed by date. The 10-K/10-Q filings are sourced from SEC EDGAR and indexed by filing date. The code confirms this: the FinMemDataset class loads data from pre-compiled pickle files (stock_data_cherrypick_2000_2024.pkl, stock_data_sp500_2000_2024.pkl), and the get_subset_by_time_range method filters data by date range. The finsaber.py code shows that subset_data is obtained via self.data_loader.get_subset_by_time_range(self.trade_config.date_from, self.trade_config.date_to), ensuring only data within the specified window is used. No live web search or API calls to mutable sources are present in the backtesting pipeline.

EVIDENCE

"Section 5: 'To prevent look-ahead bias, all data inputs are aligned with each backtest window using only information available prior to the start date.'"

"Appendix A: 'The financial news dataset, initially compiled by Dong et al. [15], comprises 15.7 million records pertaining to 4,775 S&P 500 companies, spanning the years 1999 to 2023. We have organised the news by aligning it with the respective companies and indexing it by date.'"

"Appendix A: '10-K and 10-Q filings... Each filing is indexed by company identifier and filing date to enable alignment with other datasets.'"

"Code finmem_dataset.py: get_subset_by_time_range filters data by start_date and end_date"

"Code finsaber.py: subset_data = self.data_loader.get_subset_by_time_range(self.trade_config.date_from, self.trade_config.date_to)"

Full Trace Logging

Are prompts, retrieved documents, tool outputs, and timestamps logged for temporal auditing?

✓ PASS

The code implements multiple forms of logging and trace recording. The `finsaber.py` file saves all evaluation metrics (including equity curves with timestamps) to pickle files: `'pickle.dump(eval_metrics, f)'`. Results are organized by setup name, strategy class, and date range. The code also prints progress information including selected tickers, period dates, and estimated LLM costs (via `llm_cost_monitor`). The `finsaber_bt.py` similarly saves results with `pickle.dump` and prints detailed trade information including open/close dates, prices, and PnL. The `results_aggregation.ipynb` notebook shows the structured output format. While the logging is not a comprehensive prompt-level trace system, it does save intermediate outputs, timestamps, and results to files in a structured manner sufficient for auditing.

EVIDENCE

"Code `finsaber.py`: `'pickle.dump(eval_metrics, f)'` - saves all metrics including equity_with_time DataFrames"

"Code `finsaber.py`: `'print(f\"Finish backtesting period {self.trade_config.date_from} to {self.trade_config.date_to}. Estimated cost: ${get_llm_cost()}\")'`"

"Code `finsaber.py`: `'print(f\"Selected tickers for the period {window[0]} to {window[1]}: {self.trade_config.tickers}\")'`"

"Code `finsaber_bt.py`: saves `eval_metrics` with pickle including equity curves with datetime"

"Code `finsaber_bt.py`: prints trade details including Open Date, Close Date, Price, Profit/Loss"

2. Dynamic Universe Construction Survivorship Control

Time-Indexed Tradable Universe

Does the simulation sample from the universe active at the decision time, including delisted or failed entities?

✓ PASS

The paper explicitly addresses survivorship bias as a core contribution. The data includes delisted symbols: 'our dataset includes delisted symbols that were historically part of the S&P 500 index, based on the archived constituent list' (Appendix A). The selection strategies operate on 'historical S&P 500 constituents available at the start of a given rolling-window period' (Appendix B.2). The code confirms this: the `finsaber_bt.py` file includes explicit delisting detection logic that checks if a stock's last data date is significantly before the expected end date, and if so, treats it as delisted by removing the last 7 days of data (simulating a delisting announcement period). The `stock_data_sp500_2000_2024.pkl` dataset is used for composite setups, which includes historical constituents. The selection strategies (`RandomSP500Selector`, `MomentumSP500Selector`, etc.) select from the universe available at each window start.

EVIDENCE

"Appendix A: 'our dataset includes delisted symbols that were historically part of the S&P 500 index, based on the archived constituent list. This inclusion enhances the historical completeness of our dataset and mitigates survivorship bias'"

"Section 6.2: 'To mitigate survivorship bias, we use historical constituent lists, specifically S&P 500 for US market, at each evaluation period's start and explicitly include delisted symbols.'"

"Code `finsaber_bt.py`: delisting detection - 'if `last_data_date < last_expected_date - pd.DateOffset(days=3)`: `print(f\"{ticker} appears to be delisted on {last_data_date.strftime(\"%Y-%m-%d\")}`), applying 7 days delisting announcement period.'"

"Code finsaber.py: 'status = self.framework.run(strategy, delist_check=delist_check)' with delist_check=True for rolling window setups"

Unbiased Task Sampling

Are benchmarks constructed without conditioning on ex-post news volume or present-day prominence?

✓ PASS

The paper's core contribution is explicitly addressing the bias from selective stock choices. The Composite evaluation setup uses four systematic selection strategies: Random Five (random selection from historical S&P 500 constituents), Momentum Factor, Volatility Effect, and FinCon Selection Agent. None of these condition on ex-post prominence or news volume. The Random Five strategy is explicitly described as selecting 'five stocks at random, without replacement, from the list of all available historical S&P 500 constituents for that period' (Appendix B.2). The paper evaluates across 91, 84, 63, and 80 total distinct symbols for the four selection methods respectively. The paper explicitly critiques prior work for using 'only a small number of well-known stocks such as TSLA and AMZN' and demonstrates how this inflates results. The code confirms the random selector uses a seed based on the year for reproducibility.

EVIDENCE

"Section 6.2: 'To overcome the aforementioned biases, we introduce the Composite evaluation setup within FINSABER. This setup integrates systematic selection-based strategies to expand and diversify the stock universe, explicitly addressing survivorship and data-snooping biases.'"

"Appendix B.2: 'Random Five. This strategy serves as a simple baseline... it selects five stocks at random, without replacement, from the list of all available historical S&P 500 constituents for that period.'"

"Section 4: 'Many LLM-based investing studies selectively use only a small number of well-known stocks such as TSLA and AMZN. These are both historical winners, which limits generalisability and embeds both survivorship and look-ahead bias into the evaluation.'"

"Code random_sp500_selector.py: RandomSP500Selector with random_seed_setting='year'"

Failure Regime Reporting

Does the study report outcomes separately for surviving vs. non-surviving entities?

✗ FAIL

While the paper includes delisted symbols in the universe and handles delisting in the code (finsaber_bt.py detects delisted stocks and applies a 7-day delisting announcement period), the paper does not report results stratified by survival status. There is no table or analysis showing performance separately for surviving vs. non-surviving (delisted) entities. The paper mentions including delisted symbols and reports aggregate results across all symbols in each selection strategy, but does not break down how many of the 91, 84, 63, or 80 symbols were delisted vs. surviving, nor does it show how strategy performance differs between these groups.

EVIDENCE

"The paper mentions delisted symbols are included but provides no stratified results by survival status."

"Code finsaber_bt.py handles delisting but results are aggregated without survival-status breakdown."

"Tables 3 and 4 show aggregate results across all symbols without distinguishing surviving vs. delisted entities."

RECOMMENDATIONS

Add a table or analysis showing the fraction of delisted vs. surviving symbols in each selection strategy's universe.

Report key performance metrics (Sharpe ratio, annual return, max drawdown) separately for surviving and non-surviving entities.

Discuss whether LLM strategies perform differently on delisted vs. surviving stocks, as this would strengthen the survivorship bias analysis.

3. Rationale Robustness Causal Validity

Evidence Grounding & Auditing

Are rationales explicitly grounded in retrieved documents? Are claims audited for factual hallucinations?

— N/A 

This paper is a benchmarking framework paper that evaluates LLM-based investing strategies. The LLM strategies (FinMem, FinAgent) generate trading signals (Buy/Hold/Sell), not detailed text rationales or explanations. The paper does not focus on generating or evaluating text rationales; it focuses on evaluating the trading performance of these strategies. Therefore, evidence grounding and hallucination auditing of rationales is not applicable.

EVIDENCE

"Section 3: 'Timing-Based Strategies. Timing-based strategies generate daily Buy (+1), Sell (-1), or Hold (0) signals based on market data'"

"The paper evaluates trading performance metrics (Sharpe ratio, returns, drawdown) not rationale quality"

Negative Controls & Entity Substitution

Does the system refrain from generating detailed causal stories when inputs are scrambled or entities are substituted?

— N/A 

As noted in 3.1, this paper evaluates trading signal performance, not text rationale generation. The LLM strategies produce Buy/Hold/Sell signals rather than detailed causal narratives. Entity substitution tests for narrative robustness are not applicable to this paper's methodology. The paper does perform a form of robustness testing by varying the stock universe and evaluation period, but this tests trading performance robustness, not narrative robustness.

EVIDENCE

"The paper's focus is on trading performance evaluation across different stock universes and time periods, not on rationale generation quality."

4. Epistemic Calibration Uncertainty & Abstention

Explicit Abstention Option

✓ PASS



Does the action space include 'No Trade' or 'Do Not Know', and is it scored as a legitimate outcome?

The action space explicitly includes a Hold (0) signal alongside Buy (+1) and Sell (-1). The paper states: 'Timing-based strategies generate daily Buy (+1), Sell (-1), or Hold (0) signals' (Section 3). The Hold signal effectively serves as a 'No Trade' option. The cumulative return formula in Appendix C confirms this: $C = \Pi(1 + S_t \cdot R_{m,t}) - 1$ where S_t is the position taken by the strategy on day t (+1 for long, 0 for neutral). When $S_t = 0$, the strategy abstains from trading and this is properly accounted for in the performance metrics. The paper also notes that some strategies trigger no trades at all (indicated by '-' in tables), which is reported as a legitimate outcome rather than being excluded.

EVIDENCE

"Section 3: 'Timing-Based Strategies. Timing-based strategies generate daily Buy (+1), Sell (-1), or Hold (0) signals'"

"Appendix C: 'S_t is the position taken by the strategy on day t (+1 for long, 0 for neutral)'"

"Table 2 and Table 7: '-' metrics across the board indicate no trade signals were triggered' - reported as legitimate outcome"

Calibrated Confidence/Distribution

— N/A



Does the system output a predictive distribution or a structured uncertainty estimate?

This paper evaluates timing-based trading strategies that output discrete signals (Buy/Hold/Sell), not probabilistic predictions. The evaluation framework measures trading performance through standard financial metrics (Sharpe ratio, returns, drawdown, volatility). The paper does not claim to produce calibrated probability outputs, nor is this the focus of the benchmark. The strategies are evaluated on their trading outcomes, not on probabilistic forecasting accuracy. Therefore, calibration metrics like ECE or Brier scores are not applicable.

EVIDENCE

"Section 3: Strategies generate discrete Buy (+1), Sell (-1), or Hold (0) signals"

"Appendix C: Evaluation metrics are return-based, risk-based, and risk-adjusted performance metrics, not calibration metrics"

5. Realistic Implementation Cost & Latency

Latency-Aware Execution

✗ FAIL



Are trades executed at the price available AFTER generation is complete (accounting for slippage), not the price at the start?

The paper involves backtesting trading strategies but does not discuss execution latency or slippage modeling. There is no mention of the time it takes for LLM strategies to generate signals (which could be significant given API calls to GPT-4o) or how this latency affects execution prices. The code uses BackTrader for execution, which by default executes at the next bar's open price, but there is no explicit slippage model configured. In `finsaber_bt.py`, the `cerebro` setup does not include any slippage configuration (no `cerebro.broker.set_slippage_perc()` or similar). For LLM strategies that may take seconds to minutes to generate a signal via API calls, this is a meaningful omission. The paper acknowledges transaction costs but not execution timing or slippage.

EVIDENCE

- "Code `finsaber_bt.py`: No slippage model is configured in the `cerebro` setup"
- "Code `finsaber_bt.py`: `'cerebro.broker.setcash(test_config.cash)'` and commission setup but no slippage"
- "Paper Section 6.1: 'we set a historical average risk-free rate of 0.03 and use Moomoo's standard US commission fee (\$0.0049/share, minimum \$0.99/order)' - mentions costs but not latency/slippage"

RECOMMENDATIONS

- Add a slippage model to the BackTrader configuration to account for realistic execution conditions.
- Measure and report the latency of LLM signal generation (API call time) and discuss its impact on execution prices.
- Consider executing trades at next-day open or with a delay to account for LLM inference time, especially for strategies using GPT-4o API calls.
- Add sensitivity analysis showing how results change under different slippage assumptions.

Net Utility Reporting

✓ PASS

Are metrics reported strictly after deducting transaction costs and operational inference costs?

The paper includes transaction costs in the backtesting framework. Section 6.1 states: 'we set a historical average risk-free rate of 0.03 and use Moomoo's standard US commission fee (\$0.0049/share, minimum \$0.99/order)'. The code confirms this: `finsaber_bt.py` implements a `USStockCommission` class with `commission_per_share=0.0049` and `min_commission=0.99`, and this is applied to all strategies. The `finsaber.py` also configures `commission_per_share` and `min_commission` in the framework. The behavioral analysis in Table 6 reports commission ratios, showing `FinMem`'s commission ratio is 'five to nine times higher than `FinAgent`'s'. The code also tracks LLM inference costs via `llm_cost_monitor` (`get_llm_cost()`, `reset_llm_cost()`), and prints estimated costs after each backtesting period. While LLM inference costs are tracked and printed, they are not formally deducted from returns in the reported metrics. However, transaction costs are properly deducted, and the paper reports PnL including commission in trade details.

EVIDENCE

"Section 6.1: 'we set a historical average risk-free rate of 0.03 and use Moomoo's standard US commission fee (\$0.0049/share, minimum \$0.99/order)'"

"Code finsaber_bt.py: USStockCommission class with commission_per_share=0.0049, min_commission=0.99"

"Code finsaber_bt.py: 'cerebro.broker.addcommissioninfo(commission_scheme)'"

"Code finsaber.py: 'commission_per_share=self.trade_config.__dict__.get('commission', 0.0049), min_commission=self.trade_config.__dict__.get('min_commission', 0.99)'"

"Code finsaber.py: 'print(f'Finish backtesting period... Estimated cost: \${get_llm_cost()}')'"

EXECUTIVE SUMMARY

The paper introduces FINSABER, a backtesting framework for evaluating LLM-based timing investing strategies with explicit bias mitigation. It uses rolling-window evaluations over 20 years (2004–2024) across 100+ S&P 500 symbols (including delisted stocks) to address survivorship, look-ahead, and data-snooping biases. The framework benchmarks LLM strategies (FinMem, FinAgent) against rule-based, ML/DL predictor-based, and RL-based strategies using systematic stock selection methods. Results show LLM advantages reported in prior work deteriorate under broader evaluation. Market regime analysis reveals LLM strategies are overly conservative in bull markets and overly aggressive in bear markets.

ISSUES IDENTIFIED (3)

1.1 Parametric Knowledge Cutoff Disclosure

The paper uses GPT-4o-mini and GPT-4o (as seen in Table 8: 'FinMem model=gpt-4o-mini' and 'FinAgent model=gpt-4o-mini', plus 'FinMem (GPT-4o)' results in Table 2) for LLM-based strategies evaluated over 2004–2024. However, the paper does not disclose...

- Explicitly disclose the pre-training knowledge cutoff dates for GPT-4o-mini and GPT-4o.
- Discuss the temporal overlap between the evaluation window (2004–2024) and the models' training data in a dedicated section.

2.3 Failure Regime Reporting

While the paper includes delisted symbols in the universe and handles delisting in the code (finsaber_bt.py detects delisted stocks and applies a 7-day delisting announcement period), the paper does not report results stratified by survival status. T...

- Add a table or analysis showing the fraction of delisted vs. surviving symbols in each selection strategy's universe.
- Report key performance metrics (Sharpe ratio, annual return, max drawdown) separately for surviving and non-surviving entities.

5.1 Latency-Aware Execution

The paper involves backtesting trading strategies but does not discuss execution latency or slippage modeling. There is no mention of the time it takes for LLM strategies to generate signals (which could be significant given API calls to GPT-4o) or h...

-
- Add a slippage model to the BackTrader configuration to account for realistic execution conditions.
 - Measure and report the latency of LLM signal generation (API call time) and discuss its impact on execution prices.

STRENGTHS (6)

- ✓ Point-in-Time (PIT) Retrieval
- ✓ Full Trace Logging
- ✓ Time-Indexed Tradable Universe
- ✓ Unbiased Task Sampling
- ✓ Explicit Abstention Option
- ✓ Net Utility Reporting

Not applicable to this paper: Evidence Grounding & Auditing, Negative Controls & Entity Substitution, Calibrated Confidence/Distribution.

Case 2: Forecasting

Fusing Narrative Semantics for Financial Volatility Forecasting

forecasting · anthropic/claude-sonnet-4 · 30/03/2026, 19:46:52

LLM External Data Backtest

January 2021 – December 15, 2023 (test set); training: January 2013 – December 2017; validation: January 2018 – December 2020

Time Machine GPT (TiMaGPT) TimesNet Informer Autoformer DLinear TimeXer PA4tn

FNSPID (Financial News and Stock Price Integration Dataset)

End-of-day price data (OHLCV) for KO, CMCSA, COP, GILD, MRK, NKE, ORCL VIX index data

> Paper Classification Details

1. Temporal Sanitation Non-Anticipativity

Parametric Knowledge Cutoff Disclosure

Are the latest dates of pre-training and fine-tuning disclosed? Does the evaluation window strictly follow these dates? ✓ PASS

The paper explicitly discloses the use of TiMaGPT (Time Machine GPT) as the LLM embedder and explains its temporal integrity mechanism. The paper states that 'for every date τ , the embedding $n\tau$ is generated with TiMaGPT weights whose knowledge cut-off has been rolled back to exactly one year before τ .' The code in `appendix_TMGP.py` confirms this: for each year of data, the model used is `'TiMaGPT2-{year-1}'`, e.g., for year 2013, model `'Ti-Ma/TiMaGPT2-2012'` is used. The evaluation window (test: Jan 2021 – Dec 15, 2023) strictly follows the training period (Jan 2013 – Dec 2017) and validation period (Jan 2018 – Dec 2020), with no temporal overlap. The `data_loader.py` code also confirms the strict chronological splits.

EVIDENCE

- "Paper: 'for every date τ , the embedding $n\tau$ is generated with TiMaGPT weights whose knowledge cut-off has been rolled back to exactly one year before τ '"
- "Paper: 'training span from 1 January 2013 to 31 December 2017, a validation span from 1 January 2018 to 31 December 2020, and a strictly held-out test span from 1 January 2021 to 15 December 2023'"
- "Code (`appendix_TMGP.py`): 'model_year = year - 1' and 'model_name = f'Ti-Ma/TiMaGPT2-{model_year}'"
- "Code (`data_loader.py`): 'num_train = df_raw[(df_raw[\"date\"] >= \"2013-01-01\") & (df_raw[\"date\"] <= \"2017-12-31\")].shape[0]'"

Point-in-Time (PIT) Retrieval

Are retrieval sources constructed using ONLY documents available at the specific decision time? (No live web search or mutable wikis).

✓ PASS

The paper uses the FNSPID dataset, a static, open-source financial news dataset, rather than live web search or mutable sources. The paper explicitly states that 'news documents are temporally aligned so that only articles published on the calendar day before as the corresponding price record enter the model's input for that day, thereby precluding any forward-looking leakage.' The code in `appendix_TMGPT.py` uses pre-downloaded CSV files from `./data/news_full_text/` and applies year-specific TiMaGPT models (with a one-year lag) to generate embeddings. The data_loader.py reads pre-computed embeddings from static CSV files. No live API calls or web scraping are present in the code.`

EVIDENCE

"Paper: 'News documents are temporally aligned so that only articles published on the calendar day before as the corresponding price record enter the model's input for that day, thereby precluding any forward-looking leakage.'"

"Paper: 'we extract both end-of-day price information... and the full text of associated news articles from FNSPID [14]. The FNSPID dataset [14] is an open-source resource that integrates financial news articles with corresponding time series data.'"

"Code (appendix_TMGPT.py): 'title_data = pd.read_csv(f\"./data/news_full_text/{data_values}.csv\")\""

"Code (data_loader.py): 'df_raw = pd.read_csv(os.path.join(self.root_path, self.data_path))\""

Full Trace Logging

Are prompts, retrieved documents, tool outputs, and timestamps logged for temporal auditing?

✓ PASS

The code provides meaningful logging and output saving. In `appendix_TMGPT.py`, print statements log the year and model being processed (e.g., 'Processing title for year: {year}, using model: {model_name}') and batch progress. The embeddings are saved to CSV files (e.g., `./emb_full/{data_values}_title2.csv`). In `baseline_model_result_updated_version.ipynb`, predictions are saved to CSV files in `./bench_output/` . The Step 1 and Step 2 notebooks read from results_test/` and results_test_non_vol/` directories containing metric.csv` files per run. While not a formal logging framework, the code saves intermediate outputs, predictions, and metrics to files with identifiable information, which constitutes sufficient trace logging for temporal auditing.`

EVIDENCE

"Code (appendix_TMGPT.py): 'print(f\"Processing title for year: {year}, using model: {model_name}\")\""

"Code (appendix_TMGPT.py): 'title_data.to_csv(f\"./emb_full/{data_values}_title2.csv\", index=False)\""

"Code (baseline_model_result_updated_version.ipynb): 'out_path = out_dir / f\"results_y_{ticker}_with_mom_vix_volume.csv\" and 'out_df.to_csv(out_path, index=False)\""

"Code (Step 1 Aggregate Results.ipynb): 'metrics_path = os.path.join(root_dir, fname, \"metric.csv\")\""

2. Dynamic Universe Construction Survivorship Control

Time-Indexed Tradable Universe

Does the simulation sample from the universe active at the decision time, including delisted or failed entities?

— N/A ▾

The paper uses a fixed pre-built dataset of seven large-capitalization U.S. equities (KO, CMCSA, COP, GILD, MRK, NKE, ORCL) sourced from FNSPID. The paper explicitly states these tickers were selected based on news article volume availability. This is a fixed, pre-built dataset used for volatility forecasting evaluation, not a dynamic trading simulation that requires point-in-time universe construction. The paper does not claim to simulate a tradable universe or portfolio construction process. Since the paper uses a fixed pre-built dataset, this criterion is not applicable.

EVIDENCE

"Paper: 'We construct a unified daily panel comprising seven large-capitalization U.S. equities — Coca-Cola (KO), Comcast (CMCSA), ConocoPhillips (COP), Gilead Sciences (GILD), Merck & Co. (MRK), Nike (NKE) and Oracle (ORCL).'"

"Paper: 'Owing to the limited availability of news data, above seven tickers were selected according to the volume of associated articles.'"

Unbiased Task Sampling

Are benchmarks constructed without conditioning on ex-post news volume or present-day prominence?

✗ FAIL ▾

The paper explicitly states that the seven tickers were selected based on news article volume: 'Owing to the limited availability of news data, above seven tickers were selected according to the volume of associated articles.' This selection criterion conditions on ex-post news volume, which introduces a form of selection bias — stocks with more news coverage tend to be larger, more prominent companies. All seven selected stocks (KO, CMCSA, COP, GILD, MRK, NKE, ORCL) are large-cap, well-known companies. This selection based on news volume may overrepresent high-news-volume, prominent companies and underrepresent smaller or less-covered stocks, potentially inflating the apparent benefit of news embeddings.

EVIDENCE

"Paper: 'Owing to the limited availability of news data, above seven tickers were selected according to the volume of associated articles.'"

"Paper: 'We construct a unified daily panel comprising seven large-capitalization U.S. equities — Coca-Cola (KO), Comcast (CMCSA), ConocoPhillips (COP), Gilead Sciences (GILD), Merck & Co. (MRK), Nike (NKE) and Oracle (ORCL).'"

RECOMMENDATIONS

Disclose the full set of candidate tickers considered before filtering by news volume, and report how many were excluded.

Consider including a broader, more representative sample of stocks with varying news coverage levels to test whether the news embedding benefit generalizes.

Acknowledge the potential selection bias introduced by conditioning on news volume in the paper's limitations section.

Test whether results hold for stocks with lower news coverage to assess generalizability.

Failure Regime Reporting

— N/A

Does the study report outcomes separately for surviving vs. non-surviving entities?

The paper evaluates volatility forecasting on a fixed set of seven large-cap U.S. equities that all survived throughout the evaluation period (2013–2023). The paper does not involve entity-level evaluation where some entities might fail or delist during the study period. All seven stocks (KO, CMCSA, COP, GILD, MRK, NKE, ORCL) are active, large-cap companies that remained listed throughout the entire study period. Failure regime reporting is not applicable to this methodology.

EVIDENCE

"Paper: 'We construct a unified daily panel comprising seven large-capitalization U.S. equities — Coca-Cola (KO), Comcast (CMCSA), ConocoPhillips (COP), Gilead Sciences (GILD), Merck & Co. (MRK), Nike (NKE) and Oracle (ORCL).'"

"Paper: 'Both news and time series data are partitioned chronologically into a training span from 1 January 2013 to 31 December 2017, a validation span from 1 January 2018 to 31 December 2020, and a strictly held-out test span from 1 January 2021 to 15 December 2023.'"

3. Rationale Robustness Causal Validity

Evidence Grounding & Auditing

— N/A

Are rationales explicitly grounded in retrieved documents? Are claims audited for factual hallucinations?

The paper does not generate text rationales or explanations. M2VN is a deep learning model that produces point predictions of realized volatility (a scalar output). The LLM (TiMaGPT) is used solely as an embedding model to convert news text into dense vector representations, not to generate explanatory text or rationales. The model output is $\hat{y}_{t+H} \in Y \subseteq \mathbb{R}_{\geq 0}$, a numerical volatility forecast. No text generation, citation extraction, or fact-checking pipelines are present in the code.

EVIDENCE

"Paper: 'The final component converts the fused latent sequence into a point prediction $\hat{y}_{t+H}$.'"

"Paper: 'A deterministic output is appropriate because the target y_{t+H} is defined as a smoothed aggregation of three volatility estimators.'"

"Code (data_loader.py): The model outputs are numerical predictions stored in metric.csv files, not text rationales."

Negative Controls & Entity Substitution

Does the system refrain from generating detailed causal stories when inputs are scrambled or entities are substituted?

- N/A

The paper does not generate text rationales or causal stories. M2VN produces numerical volatility forecasts, not textual explanations. Entity substitution tests and scrambled input controls are not relevant to this methodology. The paper does conduct ablation studies (removing volume or news modality), but these are quantitative performance comparisons, not tests of narrative robustness.

EVIDENCE

- "Paper: 'The final component converts the fused latent sequence into a point prediction $\hat{y}_{t+H}$.'"
- "Paper: Ablation studies in Table 2 compare numerical metrics (QLike, MAPE) without any text generation."

4. Epistemic Calibration Uncertainty & Abstention

Explicit Abstention Option

Does the action space include 'No Trade' or 'Do Not Know', and is it scored as a legitimate outcome?

- N/A

The paper is a volatility forecasting paper, not a decision-making or trading paper. M2VN produces a continuous numerical forecast of realized volatility. There is no trading action space, no buy/sell/hold decisions, and no abstention mechanism. The task is purely predictive (regression), evaluated with MAPE and QLIKE metrics. Abstention is not applicable to this regression forecasting task.

EVIDENCE

- "Paper: 'The core task is, at the close of day t , to predict the realized volatility for a future day $t+H$.'"
- "Paper: 'Model performance is quantified with Mean Absolute Percentage Error (MAPE) and the Quasi-Likelihood loss (QLIKE).'"
- "Paper: 'A deterministic output is appropriate because the target y_{t+H} is defined as a smoothed aggregation of three volatility estimators.'"

Calibrated Confidence/Distribution

Does the system output a predictive distribution or a structured uncertainty estimate?

X FAIL

The paper explicitly produces only point predictions of realized volatility, with no predictive distribution or uncertainty estimates. The paper states 'A deterministic output is appropriate because the target y_{t+H} is defined as a smoothed aggregation of three volatility estimators and therefore concentrates most of its uncertainty in the inputs rather than in the measurement.' While the paper reports standard deviations across three random seeds for the reported metrics, this is variability across runs, not a calibrated predictive uncertainty for individual forecasts. No calibration metrics (ECE, Brier score, reliability diagrams) are reported. For a volatility forecasting paper used in risk management, probabilistic outputs would be highly valuable.

EVIDENCE

"Paper: 'A deterministic output is appropriate because the target y_{t+H} is defined as a smoothed aggregation of three volatility estimators and therefore concentrates most of its uncertainty in the inputs rather than in the measurement.'"

"Paper: 'The final component converts the fused latent sequence into a point prediction $\hat{y}_{t+H}$.'"

"Code (Step 1 Aggregate Results.ipynb): Results are reported as 'mean $\pm$ std' across seeds, not as predictive distributions."

RECOMMENDATIONS

Consider outputting a predictive distribution (e.g., mean and variance) rather than a point estimate, which would be more appropriate for risk management applications.

Report calibration metrics such as prediction interval coverage or Winkler scores to assess the quality of uncertainty estimates.

Acknowledge the limitation of deterministic outputs in the context of risk management, where uncertainty quantification is critical.

Consider using conformal prediction or Bayesian approaches to provide calibrated uncertainty estimates.

5. Realistic Implementation Cost & Latency

Latency-Aware Execution

Are trades executed at the price available AFTER generation is complete (accounting for slippage), not the price at the start? — N/A

The paper is a volatility forecasting paper, not a trading strategy paper. M2VN produces volatility forecasts but does not execute trades. There is no backtesting of a trading strategy, no execution prices, and no slippage modeling. The paper explicitly frames the task as predicting realized volatility for risk management and derivative pricing purposes, not for generating trading signals. Latency-aware execution is not applicable.

EVIDENCE

"Paper: 'Forecasting financial market volatility is a critical task in quantitative finance, with broad applications in risk management, derivative pricing, and portfolio optimization.'"

"Paper: 'The core task is, at the close of day t , to predict the realized volatility for a future day $t+H$.'"

"Paper: Evaluation metrics are MAPE and QLIKE — forecasting accuracy metrics, not trading performance metrics."

Net Utility Reporting

Are metrics reported strictly after deducting transaction costs and operational inference costs? — N/A

The paper is a volatility forecasting paper that does not involve trading or cost-sensitive tasks. All reported metrics (MAPE, QLIKE) are forecasting accuracy metrics, not financial returns or utility measures. There are no transaction costs, trading fees, or API inference costs to deduct. The paper does not claim to generate trading profits. Net utility reporting is not applicable to this pure forecasting methodology.

EVIDENCE

"Paper: 'Model performance is quantified with Mean Absolute Percentage Error (MAPE) and the Quasi-Likelihood loss (QLIKE).'"

"Paper: 'Forecasting financial market volatility is a critical task in quantitative finance, with broad applications in risk management, derivative pricing, and portfolio optimization.'"

"Code (Step 1 Aggregate Results.ipynb): 'metrics.iloc[0][["mse", "qlike", "mape", "r2"]].to_dict()' — only forecasting metrics are collected."

EXECUTIVE SUMMARY

M2VN (Multi-Modal Volatility Network) fuses structured market features (OHLCV, realized volatility, VIX, momentum) with news embeddings generated by TiMaGPT—a point-in-time LLM that prevents look-ahead bias—to forecast daily realized volatility. The architecture uses a spectral-convolution backbone (TimesNet-based), gated cross-modal fusion, and an InfoNCE contrastive alignment loss to integrate the two modalities. Evaluated on seven large-cap U.S. equities against ten baselines using MAPE and QLIKE metrics, with ablation studies confirming the value of both news and volume features.

ISSUES IDENTIFIED (2)

2.2 Unbiased Task Sampling

The paper explicitly states that the seven tickers were selected based on news article volume: 'Owing to the limited availability of news data, above seven tickers were selected according to the volume of associated articles.' This selection criterio...

→ Disclose the full set of candidate tickers considered before filtering by news volume, and report how many were excluded.

→ Consider including a broader, more representative sample of stocks with varying news coverage levels to test whether the news embedding benefit generalizes.

4.2 Calibrated Confidence/Distribution

The paper explicitly produces only point predictions of realized volatility, with no predictive distribution or uncertainty estimates. The paper states 'A deterministic output is appropriate because the target y_{t+H} is defined as a smoothed aggrega...

- Consider outputting a predictive distribution (e.g., mean and variance) rather than a point estimate, which would be more appropriate for risk management applications.
- Report calibration metrics such as prediction interval coverage or Winkler scores to assess the quality of uncertainty estimates.

STRENGTHS (3)

- ✓ Parametric Knowledge Cutoff Disclosure
- ✓ Point-in-Time (PIT) Retrieval
- ✓ Full Trace Logging

Not applicable to this paper: Time-Indexed Tradable Universe, Failure Regime Reporting, Evidence Grounding & Auditing, Negative Controls & Entity Substitution, Explicit Abstention Option, Latency-Aware Execution, Net Utility Reporting.

Case 3: QA

FINDER: Financial Dataset for Evaluating RAG

benchmark · anthropic/claude-sonnet-4-6 · 30/03/2026, 23:06:47

LLM

RAG

External Data

S&P 500 companies as of December 31, 2024 (latest Form 10-K filings)

GPT-o1

Claude-3.7-Sonnet

Qwen-QWQ-32B

DeepSeek-R1-Distill-LLaMA-70B

e5-mistral-7b-instruct (E5-Mistral)

multilingual-e5-large-instruct (mE5)

gte-large-en-v1.5 (GTE)

BM25

GPT-4o-mini

GPT-4-turbo

FINDER (introduced in this paper)

FiQA

FinQA

TAT-QA

ConvFinQA

MultiHierTt

DocFinQA

FinanceBench

FinTextQA

SEC EDGAR Form 10-K filings

> Paper Classification Details

1. Temporal Sanitation Non-Anticipativity

Parametric Knowledge Cutoff Disclosure

Are the latest dates of pre-training and fine-tuning disclosed? Does the evaluation window strictly follow these dates?

X FAIL

The paper uses several LLMs for evaluation (GPT-o1, Claude-3.7-Sonnet, Qwen-QWQ-32B, DeepSeek-R1-Distill-LLaMA-70B, GPT-4o-mini, GPT-4-turbo) and embedding models (E5-Mistral, mE5, GTE). The dataset is built on S&P 500 companies as of December 31, 2024, using the latest Form 10-K filings. However, the paper does not disclose the pre-training knowledge cutoff dates for any of the LLMs used. This is particularly important because models like GPT-o1 and Claude-3.7-Sonnet may have training cutoffs that overlap with or postdate the 10-K filings used in the benchmark, potentially introducing look-ahead bias where the model's parametric knowledge already contains information from the documents being evaluated. No code was provided to verify model version strings or API call configurations.

EVIDENCE

- "The dataset covers companies from the S&P 500 index as of December 31, 2024, and is built upon two primary data sources: a document set of annual reports and a set of expert-annotated questions."
- "We evaluate the generation component of our RAG system using four state-of-the-art language models: GPT-o1 from OpenAI (Jaech et al., 2024), claude-3.7-sonnet from Anthropic, Qwen-QWQ-32B from Alibaba (Team, 2025), and deepseek-r1-distill-llama-70B from DeepSeek (Guo et al., 2025)."
- "GPT-4-turbo (Achiam et al., 2023) achieved only 9% accuracy when answering clear and straightforward questions in a closed-book setting"

RECOMMENDATIONS

Disclose the exact pre-training knowledge cutoff dates for all LLMs used (GPT-o1, Claude-3.7-Sonnet, Qwen-QWQ-32B, DeepSeek-R1-Distill-LLaMA-70B, GPT-4o-mini, GPT-4-turbo).

Explicitly verify that the evaluation window (10-K filings as of December 31, 2024) does not overlap with the training data of the models being evaluated.

Add a temporal overlap analysis section discussing whether any model's training cutoff postdates the 10-K filings used in the benchmark.

Consider using models with clearly documented cutoffs that predate the filing dates, or explicitly acknowledge and discuss the risk of parametric knowledge contamination.

Point-in-Time (PIT) Retrieval

Are retrieval sources constructed using ONLY documents available at the specific decision time? (No live web search or mutable wikis).

✓ PASS

The paper uses Form 10-K filings collected via web scraping from EDGAR as the retrieval corpus. These are static, archival documents (annual reports) that represent point-in-time snapshots of company disclosures. The paper does not use live web search, mutable wikis, or real-time data feeds. The documents were collected from EDGAR in raw HTML format and preprocessed into plain text paragraphs. Since 10-K filings are immutable once filed with the SEC, the retrieval corpus is effectively point-in-time. No code was provided to verify the exact scraping implementation, but the paper's description is consistent with PIT retrieval.

EVIDENCE

"The documents consist of the latest Form 10-K filings, which were collected via web scraping from EDGAR in raw HTML format."

"We perform basic preprocessing by converting HTML files into plain text, removing HTML tags, and segmenting the content into distinct paragraphs."

"FINDER is constructed using real-world financial inquiries from investment professionals, ensuring its relevance to industry applications."

Full Trace Logging

Are prompts, retrieved documents, tool outputs, and timestamps logged for temporal auditing?

✗ FAIL

The paper does not describe any logging infrastructure, audit trail, or trace data for prompts, retrieved documents, tool outputs, or timestamps. While the paper mentions using the RAGAS framework for evaluation and describes the experimental setup in detail, there is no mention of logging mechanisms that would allow temporal auditing of the RAG pipeline. The dataset is published on HuggingFace, but no supplementary trace data or logging artifacts are mentioned. No code was provided to verify whether logging was implemented in practice.

EVIDENCE

"To evaluate our baseline, we adopt the RAGAS framework (Es et al., 2023), which provides automated evaluation for RAG systems and integrates seamlessly with LLM-based workflows such as LANGCHAIN and LLAMAINDEX."

"Dataset is available in Huggingface."

"Prompts are kept minimal across all settings, presenting the context (if any) followed by the user query."

RECOMMENDATIONS

Implement and document a logging system that records prompts, retrieved passages, model outputs, and timestamps for each evaluation run.

Release trace logs alongside the dataset on HuggingFace to enable reproducibility and temporal auditing.

Include a description of the logging infrastructure in the paper, specifying what is captured and how it can be used for audit purposes.

Consider using structured logging frameworks (e.g., MLflow, W&B) to track experiment metadata including model versions, API call timestamps, and retrieved document IDs.

2. Dynamic Universe Construction Survivorship Control

Time-Indexed Tradable Universe

Does the simulation sample from the universe active at the decision time, including delisted or failed entities?

✗ FAIL 

The paper explicitly uses the S&P 500 index as of December 31, 2024 as its company universe. This is a static snapshot of current index constituents, which inherently introduces survivorship bias — companies that were delisted, went bankrupt, or were removed from the S&P 500 prior to December 31, 2024 are excluded from the benchmark. The paper does not discuss this limitation or attempt to include historically active but currently delisted companies. While this is a benchmark paper (not a trading simulation), the entity universe construction is still subject to survivorship bias in terms of which companies' 10-K filings are included.

EVIDENCE

"The dataset covers companies from the S&P 500 index as of December 31, 2024, and is built upon two primary data sources: a document set of annual reports and a set of expert-annotated questions."

"This filtering resulted in a final dataset comprising 5,703 questions linked to reports from 490 companies."

RECOMMENDATIONS

Acknowledge the survivorship bias introduced by using the S&P 500 as of December 31, 2024 as the company universe.

Consider including companies that were in the S&P 500 during the evaluation period but were subsequently removed or delisted.

Add a limitations section discussing how the current-constituent-only approach may skew the benchmark toward large, successful companies.

If the benchmark is intended to reflect real-world financial QA, consider expanding the universe to include mid-cap or small-cap companies, or historically active companies.

Unbiased Task Sampling

Are benchmarks constructed without conditioning on ex-post news volume or present-day prominence?

✓ PASS

The paper describes a balanced sampling approach across S&P 500 companies. Questions were sourced from a financial Q&A service database used by hedge fund analysts, portfolio managers, and investment banking analysts — representing real professional queries rather than ex-post prominent cases. The paper explicitly states that 'duplicate queries were removed and a balanced sampling across S&P 500 companies was applied.' The distribution across financial topics (company overview, financials, footnotes, governance, legal, risk, shareholder return, accounting) appears reasonably balanced. The sampling is driven by actual professional search behavior rather than ex-post news volume or present-day prominence.

EVIDENCE

"To ensure diversity and relevance, duplicate queries were removed and a balanced sampling across S&P 500 companies was applied."

"The questions were initially gathered from a financial Q&A service database used by hedge fund analysts, portfolio managers, and investment banking analysts."

"From an initial collection of 7,000 questions, we applied a rigorous filtering process: any question for which no ground truth evidence could be identified in the corresponding 10-K filing was excluded."

Failure Regime Reporting

Does the study report outcomes separately for surviving vs. non-surviving entities?

— N/A

This paper is a benchmark for financial QA using RAG systems, not an entity-level survival analysis or trading simulation. The evaluation focuses on retrieval and generation performance metrics (Context Recall, Context Precision, Response Correctness, Faithfulness) across question categories (Accounting, Company Overview, Financials, etc.), not on entity survival status. The concept of 'surviving vs. non-surviving entities' is not relevant to this benchmark's evaluation methodology.

EVIDENCE

"We first evaluate retrieval models, highlighting how neural methods outperform traditional approaches in capturing domain-specific semantics."

"Table 4 provides a comparison of retrieval performance across eight diverse financial document categories."

"Table 7 assesses baseline language models across seven tasks—Qualitative reasoning and various arithmetic operations."

3. Rationale Robustness Causal Validity

Evidence Grounding & Auditing

Are rationales explicitly grounded in retrieved documents? Are claims audited for factual hallucinations?

✓ PASS

The paper explicitly evaluates faithfulness as a core metric, which measures whether all claims in generated responses are supported by the provided context. The RAGAS framework is used to assess both Correctness and Faithfulness, where 'Faithfulness assesses consistency with the provided context. A response is considered faithful if all claims are supported by the context.' The annotation process also requires that 'Responses have to be formulated with accurate and precisely with calculations if needed, ensuring that they were both comprehensive and strictly grounded in the extracted evidence.' The benchmark is specifically designed to evaluate grounding in retrieved documents, making this a central concern of the paper.

EVIDENCE

"Faithfulness assesses consistency with the provided context. A response is considered faithful if all claims are supported by the context."

"Correctness measures factual and semantic alignment with the ground truth answer, while Faithfulness assesses consistency with the provided context."

"Answer Generation and Verification: Responses have to be formulated with accurate and precisely with calculations if needed, ensuring that they were both comprehensive and strictly grounded in the extracted evidence."

"Claude-3.7-Sonnet consistently achieves the highest Faithfulness scores, highlighting its unique ability to produce coherent and trustworthy outputs."

Negative Controls & Entity Substitution

Does the system refrain from generating detailed causal stories when inputs are scrambled or entities are substituted?

✗ FAIL

The paper does not include any negative control experiments, entity substitution tests, or scrambled input ablations. While the paper does compare performance across different context conditions (no context, retrieved context, gold context), it does not test whether models generate plausible-sounding but incorrect narratives when given scrambled or substituted entity inputs. This is particularly relevant given the paper's focus on ambiguous queries with abbreviations (e.g., 'MS' for Morgan Stanley) — a substitution test would reveal whether models hallucinate coherent stories for wrong entities. No code was provided to verify any ablation scripts.

EVIDENCE

"In the no context setting, models generate responses without any external information, simulating scenarios with no retrieval."

"FINDER uniquely assesses both the search system's ability to interpret queries (e.g., recognizing 'MS' as Morgan Stanley) and the LLM's capacity to synthesize relevant information from multiple sources."

"GPT-4-turbo (Achiam et al., 2023) achieved only 9% accuracy when answering clear and straightforward questions in a closed-book setting, with 91% of its responses being incorrect or unanswered."

RECOMMENDATIONS

Add entity substitution experiments where company names or tickers are replaced with incorrect entities to test whether models generate plausible but wrong narratives.

Include scrambled context experiments where retrieved passages are shuffled or replaced with irrelevant documents to test model robustness.

Test whether models appropriately abstain or express uncertainty when given ambiguous abbreviations that could refer to multiple entities.

Report the rate of confident but incorrect responses (hallucinations) in the no-context setting as a negative control baseline.

4. Epistemic Calibration Uncertainty & Abstention

Explicit Abstention Option

X FAIL 

Does the action space include 'No Trade' or 'Do Not Know', and is it scored as a legitimate outcome?

The paper does not include an explicit abstention option in the evaluation framework. Models are evaluated on Response Correctness and Faithfulness, but there is no mechanism for models to indicate 'I don't know' or refuse to answer when they lack sufficient information. The no-context setting shows models achieving only ~9-10% correctness, suggesting they are generating responses even when they lack the necessary information rather than abstaining. The RAGAS framework used does not appear to score abstention as a legitimate outcome. This is relevant because financial QA systems should ideally be able to express uncertainty rather than generating potentially harmful incorrect answers.

EVIDENCE

"In the no context setting, models generate responses without any external information, simulating scenarios with no retrieval."

"Without Information: Claude-3.7-Sonnet 9.37, GPT-o1 10.14, Deepseek-R1-Distill 9.88, Qwen-QWQ 9.13"

"GPT-4-turbo (Achiam et al., 2023) achieved only 9% accuracy when answering clear and straightforward questions in a closed-book setting, with 91% of its responses being incorrect or unanswered."

RECOMMENDATIONS

Add an explicit 'I don't know' or abstention option to the evaluation framework and score it as a legitimate outcome.

Measure the rate at which models abstain vs. generate incorrect answers in the no-context setting.

Design prompts that explicitly allow models to express uncertainty or refuse to answer when context is insufficient.

Report coverage (proportion of questions answered) alongside accuracy to provide a more complete picture of model performance.

Calibrated Confidence/Distribution

✗ FAIL 

Does the system output a predictive distribution or a structured uncertainty estimate?

The paper does not report any calibration metrics, probability outputs, or uncertainty estimates for the models evaluated. The evaluation relies entirely on point estimates (Response Correctness, Faithfulness, Context Recall, Context Precision) without any measure of model confidence or calibration. No reliability diagrams, ECE scores, or Brier scores are reported. The LLM-as-a-judge approach via RAGAS produces scores from 0-100 but these are not calibrated confidence estimates. Given that financial QA is a high-stakes domain where uncertainty quantification is critical, this is a notable gap.

EVIDENCE

"Both metrics range from 0 to 100, with higher values indicating better performance."

"We adopt an LLM-as-a-judge approach (Gu et al., 2024; Zheng et al., 2023; Huang et al., 2024) supported by RAGAS, enabling more flexible and context-aware evaluation of generated outputs."

"All models are used with a temperature of 0.0; other parameters remain at default settings."

RECOMMENDATIONS

Add calibration metrics such as Expected Calibration Error (ECE) or Brier scores to assess model confidence.

Extract log-probabilities from models that support it to compute calibrated confidence estimates.

Report reliability diagrams showing the relationship between model confidence and actual accuracy.

Consider evaluating models at multiple temperature settings to understand the effect of temperature on calibration.

5. Realistic Implementation Cost & Latency

Latency-Aware Execution

Are trades executed at the price available AFTER generation is complete (accounting for slippage), not the price at the start?

— N/A

This paper presents a benchmark for financial question-answering using RAG systems. It does not involve any trading, portfolio construction, or backtesting. The evaluation is purely about information retrieval and text generation quality. Latency-aware execution and slippage are not relevant concepts for this paper's methodology.

EVIDENCE

"FINDER establishes a new benchmark for financial question-answering by introducing ambiguous, domain-specific queries that reflect real-world search behavior."

"We evaluate our system on a representative 10% subset of the dataset."

Net Utility Reporting

Are metrics reported strictly after deducting transaction costs and operational inference costs?

— N/A

This paper presents a benchmark for financial question-answering using RAG systems. It does not involve trading or cost-sensitive financial decisions. While the paper does use commercial LLM APIs (GPT-o1, Claude-3.7-Sonnet, GPT-4o-mini) which incur inference costs, the paper's evaluation metrics (Context Recall, Context Precision, Response Correctness, Faithfulness) are not financial utility metrics that would require deduction of transaction or inference costs. The paper is focused on QA accuracy, not financial returns.

EVIDENCE

"FINDER establishes a new benchmark for financial question-answering by introducing ambiguous, domain-specific queries that reflect real-world search behavior."

"For analysis, we evaluate our system on a representative 10% subset of the dataset."

CROSS-CHECK INTERACTIONS

Look-Ahead + Survivorship: Temporal leakage combined with survivor-only universe creates a doubly biased evaluation.

EXECUTIVE SUMMARY

The paper introduces FINDER, a benchmark of 5,703 expert-annotated query-evidence-answer triplets derived from real-world financial queries and SEC 10-K filings for S&P 500 companies. Queries reflect professional search behavior with abbreviations and domain jargon. Baseline evaluations compare sparse (BM25) and dense retrieval models (E5-Mistral, mE5, GTE) on context recall, LLMs (GPT-o1, Claude-3.7-Sonnet, Qwen-QWQ, DeepSeek-R1) on reranking and

generation, using the RAGAS framework across no-context, retrieved-context, and gold-context settings.

ISSUES IDENTIFIED (6)

1.1 Parametric Knowledge Cutoff Disclosure

The paper uses several LLMs for evaluation (GPT-o1, Claude-3.7-Sonnet, Qwen-QWQ-32B, DeepSeek-R1-Distill-LLaMA-70B, GPT-4o-mini, GPT-4-turbo) and embedding models (E5-Mistral, mE5, GTE). The dataset is built on S&P 500 companies as of December 31, 20...

- Disclose the exact pre-training knowledge cutoff dates for all LLMs used (GPT-o1, Claude-3.7-Sonnet, Qwen-QWQ-32B, DeepSeek-R1-Distill-LLaMA-70B, GPT-4o-mini, GPT-4-turbo).
- Explicitly verify that the evaluation window (10-K filings as of December 31, 2024) does not overlap with the training data of the models being evaluated.

1.3 Full Trace Logging

The paper does not describe any logging infrastructure, audit trail, or trace data for prompts, retrieved documents, tool outputs, or timestamps. While the paper mentions using the RAGAS framework for evaluation and describes the experimental setup i...

- Implement and document a logging system that records prompts, retrieved passages, model outputs, and timestamps for each evaluation run.
- Release trace logs alongside the dataset on HuggingFace to enable reproducibility and temporal auditing.

2.1 Time-Indexed Tradable Universe

The paper explicitly uses the S&P 500 index as of December 31, 2024 as its company universe. This is a static snapshot of current index constituents, which inherently introduces survivorship bias — companies that were delisted, went bankrupt, or were...

- Acknowledge the survivorship bias introduced by using the S&P 500 as of December 31, 2024 as the company universe.
- Consider including companies that were in the S&P 500 during the evaluation period but were subsequently removed or delisted.

3.2 Negative Controls & Entity Substitution

The paper does not include any negative control experiments, entity substitution tests, or scrambled input ablations. While the paper does compare performance across different context conditions (no context, retrieved context, gold context), it does ...

- Add entity substitution experiments where company names or tickers are replaced with incorrect entities to test whether models generate plausible but wrong narratives.

→ Include scrambled context experiments where retrieved passages are shuffled or replaced with irrelevant documents to test model robustness.

4.1 Explicit Abstention Option

The paper does not include an explicit abstention option in the evaluation framework. Models are evaluated on Response Correctness and Faithfulness, but there is no mechanism for models to indicate 'I don't know' or refuse to answer when they lack su...

→ Add an explicit 'I don't know' or abstention option to the evaluation framework and score it as a legitimate outcome.

→ Measure the rate at which models abstain vs. generate incorrect answers in the no-context setting.

4.2 Calibrated Confidence/Distribution

The paper does not report any calibration metrics, probability outputs, or uncertainty estimates for the models evaluated. The evaluation relies entirely on point estimates (Response Correctness, Faithfulness, Context Recall, Context Precision) witho...

→ Add calibration metrics such as Expected Calibration Error (ECE) or Brier scores to assess model confidence.

→ Extract log-probabilities from models that support it to compute calibrated confidence estimates.

STRENGTHS (3)

- ✓ Point-in-Time (PIT) Retrieval
- ✓ Unbiased Task Sampling
- ✓ Evidence Grounding & Auditing

Not applicable to this paper: Failure Regime Reporting, Latency-Aware Execution, Net Utility Reporting.